

SOLIDWORKS API (.NET, VBA, C#) — Reference

(142/337): ILinearPatternFeatureData

SOLIDWORKS API Documentation (converted from CHM)

ILinearPatternFeatureData (Type)

SOLIDWORKS API Help

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ILinearPatternFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ILinearPatternFeatureData Interface

Allows access to a linear pattern feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ILinearPatternFeatureData
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
```

C#

```
public interface ILinearPatternFeatureData
```

C++/CLI

```
public interface class ILinearPatternFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData](#).

Example

'VBA

```
'-----  
' Preconditions: Verify that the part exists.  
,
```

```
' Postconditions:
```

```
' 1. Opens the part.
```

```
' 2. Selects the feature to pattern.
```

```
' 3. Selects the edges for Direction 1 and
```

```
' Direction 2 for the bidirectional linear
```

```
' pattern.
```

```
' 4. Creates a LPattern1.
```

```
' 5. Examine the FeatureManager design tree and
```

```
' graphics area.  
,
```

```
' NOTE: Because the part is used elsewhere, do not save  
' changes.  
'-----
```

```
Option Explicit
```

```
Dim swApp As SldWorks.SldWorks
```

```
Dim swModel As SldWorks.ModelDoc2
```

```
Dim swModelDocExt As SldWorks.ModelDocExtension
```

```
Dim swFeatureManager As SldWorks.FeatureManager
```

```
Dim swFeature As SldWorks.Feature
```

```
Dim swLinearPatternFeatureData As SldWorks.LinearPatternFeatureData
```

```
Dim status As Boolean
```

```
Dim errors As Long
```

```
Dim warnings As Long
```

```
Dim fileName As String
```

```
Sub main()
```

```
Set swApp = Application.SldWorks
```

```
fileName = "C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2019\samples\tutorial\api\box.sldprt"
```

```
Set swModel = swApp.OpenDoc6(fileName, swDocumentTypes_e.swDocPART, swOpenDocOptions_e.swOpenDocOptions_Silent, "",  
errors, warnings)
```

```
Set swModelDocExt = swModel.Extension
```

```
Set swFeatureManager = swModel.FeatureManager
```

```
'Select feature to pattern
status = swModelDocExt.SelectByID2("CBORE for #6 Binding Head Machine Screw1", "BODYFEATURE", 0, 0, 0, False, 4, Nothing, 0)
'Select edges for Direction 1 and Direction 2
status = swModelDocExt.SelectByID2("", "EDGE", -3.41223962029176E-02, 3.00321434688158E-02, 4.60303188361877E-02, True, 1
Nothing, 0)
status = swModelDocExt.SelectByID2("", "EDGE", 4.36465176948104E-02, 3.01916139486593E-02, 1.14344277122314E-02, True, 2,
Nothing, 0)
```

```
'Create linear pattern
Set swLinearPatternFeatureData = swFeatureManager.CreateDefinition(swFmLPattern)
swLinearPatternFeatureData.D1EndCondition = 0
swLinearPatternFeatureData.D1ReverseDirection = False
swLinearPatternFeatureData.D1Spacing = 0.01
swLinearPatternFeatureData.D1TotalInstances = 4
swLinearPatternFeatureData.D2EndCondition = 0
swLinearPatternFeatureData.D2PatternSeedOnly = False
swLinearPatternFeatureData.D2ReverseDirection = False
swLinearPatternFeatureData.D2Spacing = 0.01
swLinearPatternFeatureData.D2TotalInstances = 4
swLinearPatternFeatureData.GeometryPattern = False
swLinearPatternFeatureData.VarySketch = False
Set swFeature = swFeatureManager.CreateFeature(swLinearPatternFeatureData)
```

End Sub

Example

[Get Linear Pattern Feature Data \(C#\)](#)
[Get Linear Pattern Feature Data \(VB.NET\)](#)
[Get Linear Pattern Feature Data \(VBA\)](#)
[Create Bidirectional Linear Pattern \(C#\)](#)

Remarks

Read [Pattern Features and their Feature Data Objects](#).

Before calling [IFeatureManager::CreateDefinition](#), you must pre-select the entities needed to create the linear pattern feature.

If...	To select an entity, use...
Using IModelDocExtension::SelectByID2 to select entities to pattern	1 to mark Direction 1 2 to mark Direction 2 4 to mark the faces or features to pattern 256 to mark the solid bodies to pattern 134217728 to mark the structure systems to pattern 2097152 to mark up-to references 8388608 to mark seed geometry references if not using centroids to measure offsets
Directly selecting an entity	IEntity::Select4 with ISelectData::Mark= 1 to mark Direction 1 2 to mark Direction 2 IFeature::Select2 with Mark= 4 to mark the faces or features to pattern 256 to mark the bodies to pattern 134217728 to mark the structure systems to pattern 2097152 to mark up-to references 8388608 to mark seed geometry references if not using centroids to measure offsets

After calling [IFeatureManager::CreateDefinition](#), you can initialize the feature data object using:

- [ILinearPatternFeatureData::D1Axis](#) to specify Direction 1.
- [ILinearPatternFeatureData::BodyPattern](#) to specify the type of seed entities to pattern:
 - [ILinearPatternFeatureData::PatternBodyArray](#), [PatternFaceArray](#), or [PatternFeatureArray](#)
- [ILinearPatternFeatureData::D1EndCondition](#) to specify either spacing and instances or offsetting from an up-to reference:

If explicitly setting spacing and instances, use:

- [ILinearPatternFeatureData::D1Spacing](#) to specify spacing between pattern instances
- [ILinearPatternFeatureData::D1TotalInstances](#) to specify the number of pattern instances.
- [ILinearPatternFeatureData::D1ReverseDirection](#) to reverse Direction 1.

If offsetting from an up-to reference, use:

- [ILinearPatternFeatureData::D1EndReference](#) to specify the up-to reference.
- [ILinearPatternFeatureData::D1EndRefOffset](#) to specify the distance between the last pattern instance and the up-to reference.
- [ILinearPatternFeatureData::D1EndRefReverseOffset](#) to reverse the offset.

- [ILinearPatternFeatureData::D1EndUseSeedReference](#) to specify either to measure the offset from seed geometry or a centroid. If true:
 - Use [ILinearPatternFeatureData::D1EndSeedReference](#).
- [ILinearPatternFeatureData::D1EndUseSpacing](#) to specify either spacing or number of instances with the up-to reference:
 - Use [ILinearPatternFeatureData::D1Spacing](#), if [D1EndUseSpacing](#) is true.
 - Use [ILinearPatternFeatureData::D1TotalInstances](#), if [D1EndUseSpacing](#) is false.
- All of the [ILinearPatternFeatureData::D2*](#) properties to specify Direction 2 for a bidirectional linear pattern.
- Other properties such as [feature scope](#), [instances to skip](#), [propagate visual properties](#), [geometry pattern](#), and [vary sketch](#).

You must call [IFeatureManager::CreateFeature](#) after initializing this feature data object in order to successfully create the linear pattern feature.

After creating the linear pattern feature, you can:

- [vary the spacing of pattern instances](#) and dimensions by implementing [IInstanceToVaryOptions](#).
- [change the structure system to pattern](#).

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

Accessors

[IFeature::GetDefinition](#)

[IFeatureManager::CreateDefinition](#)

Access Diagram

[LinearPatternFeatureData](#)

See Also

[ILinearPatternFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalLinearPatternFeatureData Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ILinearPatternFeatureData Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : [ILinearPatternFeatureData Interface](#)

For a list of all members of this type, see [ILinearPatternFeatureData members](#).

Public Methods

Name	Description
 AccessSelections	Gains access to selections used to define the linear pattern feature.
 GetD1AxisType	Gets the type of geometry used to determine Direction 1 of this linear pattern feature.
 GetD2AxisType	Gets the type of geometry used to determine Direction 2 of this linear pattern feature.
 GetInstanceToVaryOptions	Gets the options for varying the spacing of pattern instances in this linear pattern feature.
 GetPatternBodyCount	Gets the number of seed bodies in this linear pattern feature.
 GetPatternFaceCount	Gets the number of seed faces in this linear pattern feature.
 GetPatternFeatureCount	Gets the number of seed features in this linear pattern feature.
 GetSkippedItemCount	Gets the number of instances skipped in this linear pattern feature.
 GetTransform	Gets the transform for the specified instance of this linear pattern feature.
 IAccessSelections	Obsolete. Superseded by ILinearPatternFeatureData.IAccessSelections2
 IAccessSelections2	Gains access to selections used to define the linear pattern feature.
 IGetPatternBodyArray	Gets the seed bodies for this linear pattern feature.
 IGetPatternFaceArray	Gets the seed faces in this linear pattern feature.
 IGetPatternFeatureArray	Gets the seed features used to create this linear pattern feature.
 IGetSkippedItemArray	Gets the items skipped in this linear pattern feature.
 IsDirection2Specified	Gets whether direction 2 is specified for this linear pattern feature.
 ISetPatternBodyArray	Sets the seed bodies for this linear pattern feature.
 ISetPatternFaceArray	Sets the list of patterned faces for this linear pattern feature.

	ISetPatternFeatureArray	Sets a list of pattern features for this linear pattern feature.
	ISetSkippedItemArray	Sets the list of items skipped for this linear pattern feature.
	ReleaseSelectionAccess	Releases access to the selections that created this linear pattern.
	SelfFeatureScope	Sets the feature scope, whether to autoselect the affected bodies, and the affected bodies in this linear pattern feature.
	SetInstanceToVaryOptions	Sets the options for varying the spacing of pattern instances in this circular pattern feature.

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See Also

[ILinearPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalLinearPatternFeatureData Interface](#)

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ILinearPatternFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ILinearPatternFeatureData Interface

The following tables list the members exposed by [ILinearPatternFeatureData](#).

Public Properties

Name	Description
 AutoSelect	Gets whether to automatically select all bodies in a multibody part intersected by this linear pattern feature.
 BodyPattern	Gets or sets whether to base this linear pattern feature on bodies and structure systems or features and faces.
 D1Axis	Gets or sets Direction 1 for this linear pattern feature.
 D1EndCondition	Gets or sets how to specify the spacing of pattern instances in Direction 1 of this linear pattern feature.
 D1EndReference	Gets or sets the up-to reference geometry in Direction 1 for this linear pattern feature.
 D1EndRefOffset	Gets or sets the distance of the last pattern instance from the up-to reference geometry in Direction 1 of this linear pattern feature.
 D1EndRefReverseOffset	Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 1 of this linear pattern feature.
 D1EndSeedReference	Gets or sets the pattern seed geometry to offset from the up-to-reference geometry in Direction 1 of this linear pattern feature.
 D1EndUseSeedReference	Gets or sets whether to offset a pattern seed reference or a centroid from the up-to reference geometry in Direction 1 of this linear pattern feature.
 D1EndUseSpacing	Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to reference geometry in Direction 1 for this linear pattern feature.
 D1ReverseDirection	Gets whether to reverse Direction 1 in this linear pattern feature.
 D1Spacing	Gets or sets the spacing between pattern instances in Direction 1 of this linear pattern feature.
 D1TotalInstances	Gets or sets the total number of pattern instances in Direction 1 for this linear pattern feature.
 D2Axis	Gets or sets Direction 2 for this bidirectional linear pattern feature.
 D2EndCondition	Gets or sets how to specify the spacing of pattern instances in Direction 2 of this bidirectional linear pattern feature.
 D2EndReference	Gets or sets the up-to reference geometry in Direction 2 for this bidirectional linear pattern feature.
 D2EndRefOffset	Gets or sets the distance of the last pattern instance from up-to reference geometry in Direction 2 of this bidirectional linear pattern feature.
 D2EndRefReverseOffset	Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 2 of this bidirectional linear pattern feature.
 D2EndSeedReference	Gets or sets the seed feature geometry to offset from the up-to reference geometry in Direction 2 of this bidirectional linear pattern feature.
 D2EndUseSeedReference	Gets or sets whether to offset a pattern seed reference or a centroid from the up-to reference geometry in Direction 2 of this bidirectional linear pattern feature.
 D2EndUseSpacing	Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to reference geometry in Direction 2 for this bidirectional linear pattern feature.
 D2PatternSeedOnly	Gets or sets whether to create a pattern in Direction 2 using the seed feature only, without replicating the pattern instances of Direction 1, in this bidirectional linear pattern feature.
 D2ReverseDirection	Gets whether to reverse Direction 2 in this bidirectional linear pattern feature.
 D2Spacing	Gets or sets the distance between pattern instances in Direction 2 in this bidirectional linear pattern feature.
 D2TotalInstances	Gets or sets the total number of pattern instances in Direction 2 in this bidirectional linear pattern feature.
 FeatureScope	Gets which bodies in this multibody part are affected by this linear pattern feature.

	FeatureScopeBodies	Gets the bodies in this multibody part to be affected by this linear pattern feature.
	GeometryPattern	Gets or sets whether to create the pattern using only the geometry (faces and edges) of the feature.
	InstancesToVary	Gets or sets whether to vary the spacing of pattern instances in this linear pattern feature.
	PatternBodyArray	Gets or sets the seed bodies in this linear pattern feature.
	PatternElement	Gets or sets the type of entities on which to base this linear pattern feature.
	PatternFaceArray	Gets or sets the seed faces for this linear pattern feature.
	PatternFeatureArray	Gets or sets the seed features in this linear pattern feature.
	PropagateVisualProperty	Gets or sets whether pattern instances inherit the visual properties (e.g., colors, textures, etc.) of the original seed feature in this linear pattern feature.
	SkippedItemArray	Gets or sets the items skipped in this linear pattern feature.
	StructureSystemToPatternArray	Gets or sets the structure systems to pattern in this linear pattern feature.
	VarySketch	Gets or sets whether to allow the pattern instances of a seed sketch to vary in relation to enclosing geometry in this linear pattern feature.

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Public Methods

Name	Description
	AccessSelections Gains access to selections used to define the linear pattern feature.
	GetD1AxisType Gets the type of geometry used to determine Direction 1 of this linear pattern feature.
	GetD2AxisType Gets the type of geometry used to determine Direction 2 of this linear pattern feature.
	GetInstanceToVaryOptions Gets the options for varying the spacing of pattern instances in this linear pattern feature.
	GetPatternBodyCount Gets the number of seed bodies in this linear pattern feature.
	GetPatternFaceCount Gets the number of seed faces in this linear pattern feature.
	GetPatternFeatureCount Gets the number of seed features in this linear pattern feature.
	GetSkippedItemCount Gets the number of instances skipped in this linear pattern feature.
	GetTransform Gets the transform for the specified instance of this linear pattern feature.
	IAccessSelections Obsolete. Superseded by ILinearPatternFeatureData.IAccessSelections2
	IAccessSelections2 Gains access to selections used to define the linear pattern feature.
	IGetPatternBodyArray Gets the seed bodies for this linear pattern feature.
	IGetPatternFaceArray Gets the seed faces in this linear pattern feature.
	IGetPatternFeatureArray Gets the seed features used to create this linear pattern feature.
	IGetSkippedItemArray Gets the items skipped in this linear pattern feature.
	IsDirection2Specified Gets whether direction 2 is specified for this linear pattern feature.
	ISetPatternBodyArray Sets the seed bodies for this linear pattern feature.
	ISetPatternFaceArray Sets the list of patterned faces for this linear pattern feature.
	ISetPatternFeatureArray Sets a list of pattern features for this linear pattern feature.
	ISetSkippedItemArray Sets the list of items skipped for this linear pattern feature.
	ReleaseSelectionAccess Releases access to the selections that created this linear pattern.
	SetFeatureScope Sets the feature scope, whether to autoselect the affected bodies, and the affected bodies in this linear pattern feature.
	SetInstanceToVaryOptions Sets the options for varying the spacing of pattern instances in this circular pattern feature.

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See Also

[ILinearPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalLinearPatternFeatureData Interface](#)

ILinearPatternFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ILinearPatternFeatureData Interface

For a list of all members of this type, see [ILinearPatternFeatureData members](#).

Public Properties

Name	Description
AutoSelect	Gets whether to automatically select all bodies in a multibody part intersected by this linear pattern feature.
BodyPattern	Gets or sets whether to base this linear pattern feature on bodies and structure systems or features and faces.
D1Axis	Gets or sets Direction 1 for this linear pattern feature.
D1EndCondition	Gets or sets how to specify the spacing of pattern instances in Direction 1 of this linear pattern feature.
D1EndReference	Gets or sets the up-to reference geometry in Direction 1 for this linear pattern feature.
D1EndRefOffset	Gets or sets the distance of the last pattern instance from the up-to reference geometry in Direction 1 of this linear pattern feature.
D1EndRefReverseOffset	Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 1 of this linear pattern feature.
D1EndSeedReference	Gets or sets the pattern seed geometry to offset from the up-to-reference geometry in Direction 1 of this linear pattern feature.
D1EndUseSeedReference	Gets or sets whether to offset a pattern seed reference or a centroid from the up-to reference geometry in Direction 1 of this linear pattern feature.
D1EndUseSpacing	Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to reference geometry in Direction 1 for this linear pattern feature.
D1ReverseDirection	Gets whether to reverse Direction 1 in this linear pattern feature.
D1Spacing	Gets or sets the spacing between pattern instances in Direction 1 of this linear pattern feature.
D1TotalInstances	Gets or sets the total number of pattern instances in Direction 1 for this linear pattern feature.
D2Axis	Gets or sets Direction 2 for this bidirectional linear pattern feature.
D2EndCondition	Gets or sets how to specify the spacing of pattern instances in Direction 2 of this bidirectional linear pattern feature.
D2EndReference	Gets or sets the up-to reference geometry in Direction 2 for this bidirectional linear pattern feature.
D2EndRefOffset	Gets or sets the distance of the last pattern instance from up-to reference geometry in Direction 2 of this bidirectional linear pattern feature.
D2EndRefReverseOffset	Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 2 of this bidirectional linear pattern feature.
D2EndSeedReference	Gets or sets the seed feature geometry to offset from the up-to reference geometry in Direction 2 of this bidirectional linear pattern feature.
D2EndUseSeedReference	Gets or sets whether to offset a pattern seed reference or a centroid from the up-to reference geometry in Direction 2 of this bidirectional linear pattern feature.
D2EndUseSpacing	Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to reference geometry in Direction 2 for this bidirectional linear pattern feature.
D2PatternSeedOnly	Gets or sets whether to create a pattern in Direction 2 using the seed feature only, without replicating the pattern instances of Direction 1, in this bidirectional linear pattern feature.
D2ReverseDirection	Gets whether to reverse Direction 2 in this bidirectional linear pattern feature.
D2Spacing	Gets or sets the distance between pattern instances in Direction 2 in this bidirectional linear pattern feature.
D2TotalInstances	Gets or sets the total number of pattern instances in Direction 2 in this bidirectional linear pattern feature.
FeatureScope	Gets which bodies in this multibody part are affected by this linear pattern feature.
FeatureScopeBodies	Gets the bodies in this multibody part to be affected by this linear pattern feature.
GeometryPattern	Gets or sets whether to create the pattern using only the geometry (faces and edges) of the feature.
InstancesToVary	Gets or sets whether to vary the spacing of pattern instances in this linear pattern feature.
PatternBodyArray	Gets or sets the seed bodies in this linear pattern feature.
PatternElement	Gets or sets the type of entities on which to base this linear pattern feature.
PatternFaceArray	Gets or sets the seed faces for this linear pattern feature.
PatternFeatureArray	Gets or sets the seed features in this linear pattern feature.
PropagateVisualProperty	Gets or sets whether pattern instances inherit the visual properties (e.g., colors, textures, etc.) of the original seed feature in this linear pattern feature.
SkippedItemArray	Gets or sets the items skipped in this linear pattern feature.

 StructureSystemToPatternArray	Gets or sets the structure systems to pattern in this linear pattern feature.
 VarySketch	Gets or sets whether to allow the pattern instances of a seed sketch to vary in relation to enclosing geometry in this linear pattern feature.

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See Also

[ILinearPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalLinearPatternFeatureData Interface](#)

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AccessSelections Method (ILinearPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : AccessSelections Method (ILinearPatternFeatureData)

Gains access to selections used to define the linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections( _
    ByVal TopDoc As System.Object, _
    ByVal Component As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim TopDoc As System.Object
Dim Component As System.Object
Dim value As System.Boolean

value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(
    System.object TopDoc,
    System.object Component
)
```

C++/CLI

```
System.bool AccessSelections(
    & System.Object^ TopDoc,
    & System.Object^ Component
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document (see **Remarks**)

Component

Component in which the feature is to be modified (see **Remarks**)

Return Value

True if the selections were successfully accessed, false if not

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::AccessSelections](#).

Example

[Get Linear Pattern Feature Data \(C#\)](#)
[Get Linear Pattern Feature Data \(VB.NET\)](#)
[Get Linear Pattern Feature Data \(VBA\)](#)

Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) if you modified the feature
- [ILinearPatternFeatureData::ReleaseSelectionAccess](#) if you did not

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::IAccessSelections2 Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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AutoSelect Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : AutoSelect Property (ILinearPatternFeatureData)

Gets whether to automatically select all bodies in a multibody part intersected by this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property AutoSelect As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Boolean
```

```
value = instance.AutoSelect
```

C#

```
System.bool AutoSelect {get;}
```

C++/CLI

```
property System.bool AutoSelect {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to automatically select all bodies intersected by this linear pattern feature, false to specify affected bodies

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::AutoSelect](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::GeometryPattern](#) is true.

Call [ILinearPatternFeatureData::SetFeatureScope](#) to set this property.

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

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BodyPattern Property (ILinearPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : BodyPattern Property (ILinearPatternFeatureData)

Gets or sets whether to base this linear pattern feature on bodies and structure systems or features and faces.

.NET Syntax

Visual Basic (Declaration)

```
Property BodyPattern As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.BodyPattern = value
```

```
value = instance.BodyPattern
```

C#

```
System.bool BodyPattern {get; set;}
```

C++/CLI

```
property System.bool BodyPattern {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True for bodies and structure systems, false for features and faces

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::BodyPattern](#).

Example

[Change Linear Pattern \(C#\)](#)
[Change Linear Pattern \(VB.NET\)](#)
[Change Linear Pattern \(VBA\)](#)

Remarks

If this property is:

- True, use [ILinearPatternFeatureData::PatternBodyArray](#) to specify the bodies to pattern and [ILinearPatternFeatureData::StructureSystemToPatternArray](#) to specify the structure systems to pattern.
- False, use [ILinearPatternFeatureData::PatternFaceArray](#) or [ILinearPatternFeatureData::PatternFeatureArray](#) to specify the face or feature to pattern.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

[ILinearPatternFeatureData::PatternElement Property \(\)](#)
[ILinearPatternFeatureData::IGetPatternBodyArray Method \(\)](#)
[ILinearPatternFeatureData::IGetPatternFaceArray Method \(\)](#)
[ILinearPatternFeatureData::IGetPatternFeatureArray Method \(\)](#)
[ILinearPatternFeatureData::ISetPatternBodyArray Method \(\)](#)
[ILinearPatternFeatureData::ISetPatternFaceArray Method \(\)](#)
[ILinearPatternFeatureData::ISetPatternFeatureArray Method \(\)](#)

Availability

SOLIDWORKS 2015 SP4, Revision Number 23.4

SOLIDWORKS API Help

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D1Axis Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D1Axis Property (ILinearPatternFeatureData)

Gets or sets Direction 1 for this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1Axis As System.Object

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Object
```

```
instance.D1Axis = value
```

```
value = instance.D1Axis
```

C#

```
System.Object D1Axis {get; set;}
```

C++/CLI

```
property System.Object^ D1Axis {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Direction 1 entity: linear [edge](#), [line](#), [axis](#), [dimension](#), planar [face](#), planar [surface](#), conical face, conical surface, circular edge, or reference [plane](#)

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D1Axis](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetD1AxisType Method](#)
[ILinearPatternFeatureData::D1ReverseDirection Property](#)
[ILinearPatternFeatureData::D1Spacing Property](#)
[ILinearPatternFeatureData::D1TotalInstances Property](#)
[ILinearPatternFeatureData::VarySketch Property \(\)](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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D1EndCondition Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D1EndCondition Property (ILinearPatternFeatureData)

Gets or sets how to specify the spacing of pattern instances in Direction 1 of this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1EndCondition As System.Integer

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Integer
```

```
instance.D1EndCondition = value
```

```
value = instance.D1EndCondition
```

C#

```
System.int D1EndCondition {get; set;}
```

C++/CLI

```
property System.int D1EndCondition {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Spacing of pattern instances in Direction 1 as defined in [swPatternEndCondition_e](#)

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D1EndCondition](#).

Example

See the [ILinearPatternFeatureData](#) example.

Remarks

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1EndReference Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D1EndReference Property (ILinearPatternFeatureData)

Gets or sets the up-to reference geometry in Direction 1 for this linear pattern feature.

.NET Syntax

```
Property D1EndReference As System.Object
```



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Object

instance.D1EndReference = value

value = instance.D1EndReference
```

```
System.Object D1EndReference {get; set;}
```

```
property System.Object^ D1EndReference {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pattern's up-to reference geometry ([vertex](#), [edge](#), [face](#), or [plane](#)) in Direction 1

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D1EndReference](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

To control the pattern in Direction 1, also set:

- [ILinearPatternFeatureData::D1Axis](#)
- [ILinearPatternFeatureData::D1EndRefOffset](#)
- [ILinearPatternFeatureData::D1EndRefReverseOffset](#)
- [ILinearPatternFeatureData::D1EndUseSeedReference](#)
 - [ILinearPatternFeatureData::D1EndSeedReference](#) (if [D1EndUseSeedReference](#) is true)
- [ILinearPatternFeatureData::D1EndUseSpacing](#)
 - [ILinearPatternFeatureData::D1Spacing](#) (if [D1EndUseSpacing](#) is true)
 - [ILinearPatternFeatureData::D1TotalInstances](#) (if [D1EndUseSpacing](#) is false)

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

D1EndRefOffset Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D1EndRefOffset Property (ILinearPatternFeatureData)

Gets or sets the distance of the last pattern instance from the up-to reference geometry in Direction 1 of this linear pattern feature.

.NET Syntax

Property D1EndRefOffset As System.Double

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Double

instance.D1EndRefOffset = value

value = instance.D1EndRefOffset
```

C#

```
System.double D1EndRefOffset {get; set;}
```

C++/CLI

```
property System.double D1EndRefOffset {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Offset of last pattern instance from [ILinearPatternFeatureData::D1EndReference](#)

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::D1EndRefOffset](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

Whether to offset instance geometry or a centroid is governed by [ILinearPatternFeatureData::D1EndUseSeedReference](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1EndRefReverseOffset Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D1EndRefReverseOffset Property (ILinearPatternFeatureData)

Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 1 of this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1EndRefReverseOffset As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Boolean

instance.D1EndRefReverseOffset = value
```

```
value = instance.D1EndRefReverseOffset
```

C#

```
System.bool D1EndRefReverseOffset {get; set;}
```

C++/CLI

```
property System.bool D1EndRefReverseOffset {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse [ILinearPatternFeatureData::D1EndRefOffset](#) from [ILinearPatternFeatureData::D1EndReference](#), false to not

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D1EndRefReverseOffset](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1EndSeedReference Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D1EndSeedReference Property (ILinearPatternFeatureData)

Gets or sets the pattern seed geometry to offset from the up-to-reference geometry in Direction 1 of this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D1EndSeedReference As System.Object
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Object
```

```
instance.D1EndSeedReference = value
```

```
value = instance.D1EndSeedReference
```

C#

```
System.object D1EndSeedReference {get; set;}
```

C++/CLI

```
property System.Object^ D1EndSeedReference {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Seed geometry ([vertex](#), [edge](#), [face](#), or [plane](#)) to offset from [ILinearPatternFeatureData::D1EndReference](#)

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::D1EndSeedReference](#).

Remarks

This property is valid only if:

- [ILinearPatternFeatureData::D1EndUseSeedReference](#) is set to true
- and -
- [ILinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1EndUseSeedReference Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D1EndUseSeedReference Property (ILinearPatternFeatureData)

Gets or sets whether to offset a pattern seed reference or a centroid from the up-to reference geometry in Direction 1 of this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1EndUseSeedReference As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Boolean
```

```
instance.D1EndUseSeedReference = value
```

```
value = instance.D1EndUseSeedReference
```

C#

```
System.bool D1EndUseSeedReference {get; set;}
```

C++/CLI

```
property System.bool D1EndUseSeedReference {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to offset a pattern seed reference, false to offset a centroid

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::D1EndUseSeedReference](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

If this property is set to true, use [ILinearPatternFeatureData::D1EndSeedReference](#) to specify the seed geometry to offset from [ILinearPatternFeatureData::D1EndReference](#).

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1EndUseSpacing Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D1EndUseSpacing Property (ILinearPatternFeatureData)

Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to reference geometry in Direction 1 for this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1EndUseSpacing As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D1EndUseSpacing = value
```

```
value = instance.D1EndUseSpacing
```

C#

```
System.bool D1EndUseSpacing {get; set;}
```

C++/CLI

```
property System.bool D1EndUseSpacing {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use spacing, false to use number of pattern instances in Direction 1

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D1EndUseSpacing](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1ReverseDirection Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D1ReverseDirection Property (ILinearPatternFeatureData)

Gets whether to reverse Direction 1 in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1ReverseDirection As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D1ReverseDirection = value
```

```
value = instance.D1ReverseDirection
```

C#

```
System.bool D1ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool D1ReverseDirection {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse Direction 1, false to not

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D1ReverseDirection](#).

Example

See the [ILinearPatternFeatureData](#) example.

Remarks

This property is valid only if [ILinearPatternFeatureData::D1EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_SpacingAndInstances](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearFeaturePatternData::GetD1AxisType Method](#)
[ILinearFeaturePatternData::D1Axis Property](#)
[ILinearFeaturePatternData::D1Spacing Property](#)
[ILinearFeaturePatternData::D1TotalInstances Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1Spacing Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D1Spacing Property (ILinearPatternFeatureData)

Gets or sets the spacing between pattern instances in Direction 1 of this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1Spacing As System.Double

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Double
```

```
instance.D1Spacing = value
```

```
value = instance.D1Spacing
```

C#

```
System.double D1Spacing {get; set;}
```

C++/CLI

```
property System.double D1Spacing {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance in meters between pattern instances in Direction 1

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D1Spacing](#).

Example

See the [ILinearPatternFeatureData](#) example.

Remarks

You can set this property when [ILinearPatternFeatureData::D1EndCondition](#) is set to:

- [swPatternEndCondition_e.swPatternEndCondition_SpacingAndInstances](#)
- or -
- [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#) and [ILinearPatternFeatureData::D1EndUseSpacing](#) is set to true.

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetD1AxisType Method](#)
[ILinearPatternFeatureData::D1Axis Property](#)
[ILinearPatternFeatureData::D1ReverseDirection Property](#)
[ILinearPatternFeatureData::D1TotalInstances Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1TotalInstances Property (ILinearPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D1TotalInstances Property (ILinearPatternFeatureData)

Gets or sets the total number of pattern instances in Direction 1 for this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D1TotalInstances As System.Integer

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Integer
```

```
instance.D1TotalInstances = value
```

```
value = instance.D1TotalInstances
```

C#

```
System.int D1TotalInstances {get; set;}
```

C++/CLI

```
property System.int D1TotalInstances {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Total number of pattern instances in Direction 1

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D1TotalInstances](#).

Example

See the [ILinearPatternFeatureData](#) example.

Example

[Get Linear Pattern Feature Data \(C#\)](#)
[Get Linear Pattern Feature Data \(VB.NET\)](#)
[Get Linear Pattern Feature Data \(VBA\)](#)

Remarks

You can set this property when [ILinearPatternFeatureData::D1EndCondition](#) is set to:

- [swPatternEndCondition_e.swPatternEndCondition_SpacingAndInstances](#)
- OR -
- [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#) and [ILinearPatternFeatureData::D1EndUseSpacing](#) is set to false.

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetD1AxisType Method](#)
[ILinearPatternFeatureData::D1Axis Property](#)
[ILinearPatternFeatureData::D1ReverseDirection Property](#)
[ILinearPatternFeatureData::D1Spacing Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2Axis Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2Axis Property (ILinearPatternFeatureData)

Gets or sets Direction 2 for this bidirectional linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2Axis As System.Object

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Object
```

```
instance.D2Axis = value
```

```
value = instance.D2Axis
```

C#

```
System.object D2Axis {get; set;}
```

C++/CLI

```
property System.Object^ D2Axis {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Direction 2 entity: linear [edge](#), [line](#), [axis](#), [dimension](#), planar [face](#), planar [surface](#), conical face, conical surface, circular edge, or reference [plane](#)

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D2Axis](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetD2AxisType Method](#)
[ILinearPatternFeatureData::IsDirection2Specified Method](#)
[ILinearPatternFeatureData::D2PatternSeedOnly Property](#)
[ILinearPatternFeatureData::D2ReverseDirection Property](#)
[ILinearPatternFeatureData::D2Spacing Property](#)
[ILinearPatternFeatureData::D2TotalInstances Property](#)

Availability

D2EndCondition Property (ILinearPatternFeatureData)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2EndCondition Property (ILinearPatternFeatureData)

Gets or sets how to specify the spacing of pattern instances in Direction 2 of this bidirectional linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2EndCondition As System.Integer

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Integer
```

```
instance.D2EndCondition = value
```

```
value = instance.D2EndCondition
```

C#

```
System.int D2EndCondition {get; set;}
```

C++/CLI

```
property System.int D2EndCondition {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).**Property Value**Spacing of pattern instances in Direction 2 as defined in [swPatternEndCondition_e](#)

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D2EndCondition](#).

Example

See the [ILinearPatternFeatureData](#) example.

Remarks

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

D2EndReference Property (ILinearPatternFeatureData)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2EndReference Property (ILinearPatternFeatureData)

Gets or sets the up-to reference geometry in Direction 2 for this bidirectional linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2EndReference As System.Object

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Object

instance.D2EndReference = value

value = instance.D2EndReference
```

C#

```
System.object D2EndReference {get; set;}
```

C++/CLI

```
property System.Object^ D2EndReference {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pattern's up-to reference ([vertex](#), [edge](#), [face](#), or [plane](#)) in Direction 2

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D2EndReference](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

D2EndRefOffset Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2EndRefOffset Property (ILinearPatternFeatureData)

Gets or sets the distance of the last pattern instance from up-to reference geometry in Direction 2 of this bidirectional linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2EndRefOffset As System.Double

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Double

instance.D2EndRefOffset = value

value = instance.D2EndRefOffset
```

C#

```
System.double D2EndRefOffset {get; set;}
```

C++/CLI

```
property System.double D2EndRefOffset {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Offset of the last pattern instance from [ILinearPatternFeatureData::D2EndReference](#)

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::D2EndRefOffset](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

Whether to offset instance geometry or a centroid is governed by [ILinearPatternFeatureData::D2EndUseSeedReference](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

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SOLIDWORKS API Help

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D2EndRefReverseOffset Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2EndRefReverseOffset Property (ILinearPatternFeatureData)

Gets or sets whether to reverse the offset from the up-to-reference geometry in Direction 2 of this bidirectional linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D2EndRefReverseOffset As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Boolean

instance.D2EndRefReverseOffset = value

value = instance.D2EndRefReverseOffset
```

C#

```
System.bool D2EndRefReverseOffset {get; set;}
```

C++/CLI

```
property System.bool D2EndRefReverseOffset {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse [ILinearPatternFeatureData::D2EndRefOffset](#) from [ILinearPatternFeatureData::D2EndReference](#), false to not

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D2EndRefReverseOffset](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2EndSeedReference Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2EndSeedReference Property (ILinearPatternFeatureData)

Gets or sets the seed feature geometry to offset from the up-to reference geometry in Direction 2 of this bidirectional linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D2EndSeedReference As System.Object
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Object
```

```
instance.D2EndSeedReference = value
```

```
value = instance.D2EndSeedReference
```

C#

```
System.object D2EndSeedReference {get; set;}
```

C++/CLI

```
property System.Object^ D2EndSeedReference {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Seed geometry ([vertex](#), [edge](#), [face](#), or [plane](#)) to offset from [ILinearPatternFeatureData::D2EndReference](#)

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D2EndSeedReference](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

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D2EndUseSeedReference Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2EndUseSeedReference Property (ILinearPatternFeatureData)

Gets or sets whether to offset a pattern seed reference or a centroid from the up-to reference geometry in Direction 2 of this bidirectional linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2EndUseSeedReference As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D2EndUseSeedReference = value
```

```
value = instance.D2EndUseSeedReference
```

C#

```
System.bool D2EndUseSeedReference {get; set;}
```

C++/CLI

```
property System.bool D2EndUseSeedReference {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to offset a pattern seed reference, false to offset a centroid

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D2EndUseSeedReference](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

If this property is set to true, use [ILinearPatternFeatureData::D2EndSeedReference](#) to specify the seed geometry to offset from [ILinearPatternFeatureData::D2EndReference](#).

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2EndUseSpacing Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2EndUseSpacing Property (ILinearPatternFeatureData)

Gets or sets whether to use spacing or a number of pattern instances when offsetting up-to reference geometry in Direction 2 for this bidirectional linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D2EndUseSpacing As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D2EndUseSpacing = value
```

```
value = instance.D2EndUseSpacing
```

C#

```
System.bool D2EndUseSpacing {get; set;}
```

C++/CLI

```
property System.bool D2EndUseSpacing {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use spacing, false to use number of pattern instances in Direction 2

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D2EndUseSpacing](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2PatternSeedOnly Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2PatternSeedOnly Property (ILinearPatternFeatureData)

Gets or sets whether to create a pattern in Direction 2 using the seed feature only, without replicating the pattern instances of Direction 1, in this bidirectional linear pattern feature.

Visual Basic Syntax

Visual Basic (Declaration)

Property D2PatternSeedOnly As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D2PatternSeedOnly = value
```

```
value = instance.D2PatternSeedOnly
```

C#

```
System.bool D2PatternSeedOnly {get; set;}
```

C++/CLI

```
property System.bool D2PatternSeedOnly {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to create a pattern in Direction 2 using the seed feature only, false to create a pattern in Direction 2 using the seed feature and all of the instances generated in Direction 1

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D2PatternSeedOnly](#).

Example

See the [ILinearPatternFeatureData](#) example.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetD2AxisType Method](#)
[ILinearPatternFeatureData::IsDirection2Specified Method](#)
[ILinearPatternFeatureData::D2Axis Property](#)
[ILinearPatternFeatureData::D2ReverseDirection Property](#)
[ILinearPatternFeatureData::D2Spacing Property](#)
[ILinearPatternFeatureData::D2TotalInstances Property](#)

Availability

SOLIDWORKS 2001Plus SP2, Revision Number 10.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2ReverseDirection Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2ReverseDirection Property (ILinearPatternFeatureData)

Gets whether to reverse Direction 2 in this bidirectional linear pattern feature.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property D2ReverseDirection As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Boolean
```

```
instance.D2ReverseDirection = value
```

```
value = instance.D2ReverseDirection
```

C#

```
System.bool D2ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool D2ReverseDirection {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse Direction 2, false to not

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D2ReverseDirection](#).

Example

See the [ILinearPatternFeatureData](#) example.

Remarks

This property is valid only if [ILinearPatternFeatureData::D2EndCondition](#) is set to [swPatternEndCondition_e.swPatternEndCondition_SpacingAndInstances](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetD2AxisType Method](#)
[ILinearPatternFeatureData::IsDirection2Specified Method](#)
[ILinearPatternFeatureData::D2Axis Property](#)
[ILinearPatternFeatureData::D2Spacing Property](#)
[ILinearPatternFeatureData::D2TotalInstances Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2Spacing Property (ILinearPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2Spacing Property (ILinearPatternFeatureData)

Gets or sets the distance between pattern instances in Direction 2 in this bidirectional linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2Spacing As System.Double

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Double

instance.D2Spacing = value
value = instance.D2Spacing
```

C#

```
System.double D2Spacing {get; set;}
```

C++/CLI

```
property System.double D2Spacing {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance in meters between pattern instances in Direction 2

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D2Spacing](#).

Example

See the [ILinearPatternFeatureData](#) example.

Example

[Get Linear Pattern Feature Data \(C#\)](#)
[Get Linear Pattern Feature Data \(VB.NET\)](#)
[Get Linear Pattern Feature Data \(VBA\)](#)

Remarks

You can set this property when [ILinearPatternFeatureData::D2EndCondition](#) is set to:

- [swPatternEndCondition_e.swPatternEndCondition_SpacingAndInstances](#)
- OR -
- swPatternEndCondition_e.swPatternEndCondition_UpToReference and [ILinearPatternFeatureData::D2EndUseSpacing](#) is set to true.

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetD2AxisType Method](#)
[ILinearPatternFeatureData::IsDirection2Specified Method](#)
[ILinearPatternFeatureData::D2Axis Property](#)
[ILinearPatternFeatureData::D2PatternSeedOnly Property](#)
[ILinearPatternFeatureData::D2ReverseDirection Property](#)
[ILinearPatternFeatureData::D2TotalInstances Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

D2TotalInstances Property (ILinearPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : D2TotalInstances Property (ILinearPatternFeatureData)

Gets or sets the total number of pattern instances in Direction 2 in this bidirectional linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2TotalInstances As System.Integer

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Integer
```

```
instance.D2TotalInstances = value
```

```
value = instance.D2TotalInstances
```

C#

```
System.int D2TotalInstances {get; set;}
```

C++/CLI

```
property System.int D2TotalInstances {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Total number of pattern instances in Direction 2

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::D2TotalInstances](#).

Example

See the [ILinearPatternFeatureData](#) example.

Example

[Get Linear Pattern Feature Data \(C#\)](#)
[Get Linear Pattern Feature Data \(VB.NET\)](#)
[Get Linear Pattern Feature Data \(VBA\)](#)

Remarks

You can set this property when [ILinearPatternFeatureData::D2EndCondition](#) is set to:

- [swPatternEndCondition_e.swPatternEndCondition_SpacingAndInstances](#)
- Or -
- [swPatternEndCondition_e.swPatternEndCondition_UpToReference](#) and [ILinearPatternFeatureData::D2EndUseSpacing](#) is set to false.

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetD2AxisType Method](#)
[ILinearPatternFeatureData::IsDirection2Specified Method](#)
[ILinearPatternFeatureData::D2Axis Property](#)

[ILinearPatternFeatureData::D2PatternSeedOnly Property](#)
[ILinearPatternFeatureData::D2ReverseDirection Property](#)
[ILinearPatternFeatureData::D2Spacing Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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FeatureScope Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : FeatureScope Property (ILinearPatternFeatureData)

Gets which bodies in this multibody part are affected by this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property FeatureScope As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Boolean
```

```
value = instance.FeatureScope
```

C#

```
System.bool FeatureScope {get;}
```

C++/CLI

```
property System.bool FeatureScope {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to specify affected bodies, false to apply the pattern to all bodies every time the feature regenerates

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::FeatureScope](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::GeometryPattern](#) is true.

Call [ILinearPatternFeatureData::SetFeatureScope](#) to set this property.

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureScopeBodies Property (ILinearPatternFeatureData)

[See Also](#)



Gets the bodies in this multibody part to be affected by this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property FeatureScopeBodies As System.Object
```

Visual Basic (Usage)

```
Dim instance As ILinearPatternFeatureData
Dim value As System.Object

value = instance.FeatureScopeBodies
```

C#

```
System.Object FeatureScopeBodies {get;}
```

C++/CLI

```
property System.Object^ FeatureScopeBodies {
    System.Object^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [bodies](#) to be affected

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::FeatureScopeBodies](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::GeometryPattern](#) is true.

Call [ILinearPatternFeatureData::SetFeatureScope](#) to set this property.

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

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GeometryPattern Property (ILinearPatternFeatureData)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > ILinearPatternFeatureData Interface : GeometryPattern Property (ILinearPatternFeatureData)

Gets or sets whether to create the pattern using only the geometry (faces and edges) of the feature.

.NET Syntax

Visual Basic (Declaration)

```
Property GeometryPattern As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Boolean

instance.GeometryPattern = value

value = instance.GeometryPattern
```

C#

```
System.bool GeometryPattern {get; set;}
```

C++/CLI

```
property System.bool GeometryPattern {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use only the geometry of the feature to create the pattern (faster), false to pattern and solve each instance of the feature in its entirety (slower)

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::GeometryPattern](#).

Example

See the [ILinearPatternFeatureData](#) example.

Remarks

This property is valid only if [ILinearPatternFeatureData::BodyPattern](#) is false.

If this property is set to true, you can call [ILinearPatternFeatureData::SetFeatureScope](#).

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetD1AxisType Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : GetD1AxisType Method (ILinearPatternFeatureData)

Gets the type of geometry used to determine Direction 1 of this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetD1AxisType() As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ILinearPatternFeatureData
Dim value As System.Integer
```



Copy Code

```
value = instance.GetD1AxisType()
```

C#

```
System.int GetD1AxisType()
```

C++/CLI

```
System.int GetD1AxisType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Geometry type used to create the linear pattern in Direction 1:

- 0 = axis or ref axis
- 1 = edge
- 2 = dimension
- 3 = sketch segment

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::GetD1AxisType](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::D1ReverseDirection Property](#)
[ILinearPatternFeatureData::D1Spacing Property](#)
[ILinearPatternFeatureData::D1TotalInstances Property](#)
[ILinearPatternFeatureData::GetD2AxisType Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetD2AxisType Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : GetD2AxisType Method (ILinearPatternFeatureData)

Gets the type of geometry used to determine Direction 2 of this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetD2AxisType() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.GetD2AxisType()
```

C#

```
System.int GetD2AxisType()
```

C++/CLI

```
System.int GetD2AxisType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Geometry type used to create the linear pattern in Direction 2:

- 0 = axis or ref axis
- 1 = edge
- 2 = dimension
- 3 = sketch segment

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::GetD2AxisType](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::D2PatternSeedOnly Property](#)
[ILinearPatternFeatureData::D2ReverseDirection Property](#)
[ILinearPatternFeatureData::D2Spacing Property](#)
[ILinearPatternFeatureData::D2TotalInstances Property](#)
[ILinearPatternFeatureData::GetD1AxisType Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetInstanceToVaryOptions Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : GetInstanceToVaryOptions Method (ILinearPatternFeatureData)

Gets the options for varying the spacing of pattern instances in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetInstanceToVaryOptions() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Object  
  
value = instance.GetInstanceToVaryOptions()
```

C#

```
System.Object GetInstanceToVaryOptions()
```

C++/CLI

```
System.Object^ GetInstanceToVaryOptions();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IInstanceToVaryOptions](#)

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::GetInstanceToVaryOptions](#).

Example

See the [IInstanceToVaryOptions](#) examples.

Remarks

This method is valid only if [ILinearPatternFeatureData::InstancesToVary](#) is true.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::SetInstanceToVaryOptions Method \(\)](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPatternBodyCount Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : GetPatternBodyCount Method (ILinearPatternFeatureData)

Gets the number of seed bodies in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPatternBodyCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.GetPatternBodyCount()
```

C#

```
System.int GetPatternBodyCount()
```

C++/CLI

```
System.int GetPatternBodyCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of seed bodies

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::GetPatternBodyCount](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::IGetPatternBodyArray Method](#)
[ILinearPatternFeatureData::PatternBodyArray Property](#)
[ILinearPatternFeatureData::ISetPatternBodyArray Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPatternFaceCount Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : GetPatternFaceCount Method (ILinearPatternFeatureData)

Gets the number of seed faces in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPatternFaceCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.GetPatternFaceCount()
```

C#

```
System.int GetPatternFaceCount()
```

C++/CLI

```
System.int GetPatternFaceCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of seed faces

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::GetPatternFaceCount](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::IGetPatternFaceArray Method](#)
[ILinearPatternFeatureData::ISetPatternFaceArray Method](#)
[ILinearPatternFeatureData::PatternFaceArray Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

[SOLIDWORKS API Help](#)

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GetPatternFeatureCount Method (ILinearPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : GetPatternFeatureCount Method (ILinearPatternFeatureData)

Gets the number of seed features in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPatternFeatureCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.GetPatternFeatureCount()
```

C#

```
System.int GetPatternFeatureCount()
```

C++/CLI

```
System.int GetPatternFeatureCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of seed features

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::GetPatternFeatureCount](#).

Example

[Get Linear Pattern Feature Data \(C#\)](#)
[Get Linear Pattern Feature Data \(VB.NET\)](#)
[Get Linear Pattern Feature Data \(VBA\)](#)

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::IGetPatternFeatureArray Method](#)
[ILinearPatternFeatureData::ISetPatternFeatureArray Method](#)
[ILinearPatternFeatureData::PatternFeatureArray Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSkippedItemCount Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : GetSkippedItemCount Method (ILinearPatternFeatureData)

Gets the number of instances skipped in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSkippedItemCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.GetSkippedItemCount()
```

C#

```
System.int GetSkippedItemCount()
```

C++/CLI

```
System.int GetSkippedItemCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of skipped instances

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::GetSkippedItemCount](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::IGetSkippedItemArray Method](#)
[ILinearPatternFeatureData::ISetSkippedItemArray Method](#)
[ILinearPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetTransform Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : GetTransform Method (ILinearPatternFeatureData)

Gets the transform for the specified instance of this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTransform( _  
    ByVal Instance As System.Integer _  
) As MathTransform
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim Instance As System.Integer  
Dim value As MathTransform  
  
value = instance.GetTransform(Instance)
```

C#

```
MathTransform GetTransform(  
    System.int Instance  
)
```

C++/CLI

```
MathTransform^ GetTransform(  
& System.int Instance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance

Index of one repeating element in this pattern (see **Remarks**)

Return Value

[Transform](#)

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::GetTransform](#).

Remarks

1 <= *Instance* <= ([ILinearPatternFeatureData::D1TotalInstances](#) + [ILinearPatternFeatureData::D2TotalInstances](#))

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IAccessSelections Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : IAccessSelections Method (ILinearPatternFeatureData)

Obsolete. Superseded by [ILinearPatternFeatureData::IAccessSelections2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IAccessSelections( _  
    ByVal TopDoc As ModelDoc, _  
    ByVal Component As Component _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim TopDoc As ModelDoc  
Dim Component As Component  
Dim value As System.Boolean  
  
value = instance.IAccessSelections(TopDoc, Component)
```

C#

```
System.bool IAccessSelections(  
    ModelDoc TopDoc,  
    Component Component  
)
```

C++/CLI

```
System.bool IAccessSelections(  
& ModelDoc^ TopDoc,  
& Component^ Component  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Component

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::IAccessSelections](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

SOLIDWORKS API Help

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IAccessSelections2 Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : IAccessSelections2 Method (ILinearPatternFeatureData)

Gains access to selections used to define the linear pattern feature.

 .NET Syntax

Visual Basic (Declaration)

```
Function IAccessSelections2( _  
    ByVal TopDoc As ModelDoc2, _  
    ByVal Component As Component2 _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim TopDoc As ModelDoc2  
Dim Component As Component2  
Dim value As System.Boolean  
  
value = instance.IAccessSelections2(TopDoc, Component)
```

C#

```
System.bool IAccessSelections2(  
    ModelDoc2 TopDoc,  
    Component2 Component  
)
```

C++/CLI

```
System.bool IAccessSelections2(  
& ModelDoc2^ TopDoc,  
& Component2^ Component  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document (see **Remarks**)

Component

Component in which the feature is to be modified (see **Remarks**)

Return Value

True if the selections were successfully accessed, false if not

 Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::IAccessSelections2](#).

 Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::IModifyDefinition2](#) if you modified the feature
- [ILinearPatternFeatureData::ReleaseSelectionAccess](#) if you did not

 See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::AccessSelections Method](#)

 Availability

IGetPatternBodyArray Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : IGetPatternBodyArray Method (ILinearPatternFeatureData)

Gets the seed bodies for this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPatternBodyArray() As Body2
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As Body2
```

```
value = instance.IGetPatternBodyArray()
```

C#

```
Body2 IGetPatternBodyArray()
```

C++/CLI

```
Body2^ IGetPatternBodyArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of seed [bodies](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetPatternBodyCount Method](#)
[ILinearPatternFeatureData::ISetPatternBodyArray Method](#)
[ILinearPatternFeatureData::PatternBodyArray Property](#)
[ILinearPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

IGetPatternFaceArray Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : IGetPatternFaceArray Method (ILinearPatternFeatureData)

Gets the seed faces in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPatternFaceArray() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Object

value = instance.IGetPatternFaceArray()
```

C#

```
System.Object IGetPatternFaceArray()
```

C++/CLI

```
System.Object^ IGetPatternFaceArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of seed [faces](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetPatternFaceCount Method](#)
[ILinearPatternFeatureData::ISetPatternFaceArray Method](#)
[ILinearPatternFeatureData::PatternFaceArray Property](#)
[ILinearPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPatternFeatureArray Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : IGetPatternFeatureArray Method (ILinearPatternFeatureData)

Gets the seed features used to create this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPatternFeatureArray() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Object

value = instance.IGetPatternFeatureArray()
```

C#

```
System.Object IGetPatternFeatureArray()
```

C++/CLI

```
System.Object^ IGetPatternFeatureArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to the array of features used to create this linear pattern
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetPatternFeatureCount Method](#)
[ILinearPatternFeatureData::ISetPatternFeatureArray Method](#)
[ILinearPatternFeatureData::PatternFeatureArray Property](#)
[ILinearPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSkippedItemArray Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : IGetSkippedItemArray Method (ILinearPatternFeatureData)

Gets the items skipped in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSkippedItemArray() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Integer

value = instance.IGetSkippedItemArray()
```

C#

```
System.int IGetSkippedItemArray()
```

C++/CLI

```
System.int IGetSkippedItemArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of the skipped items
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

This array is 0-based, so if you skipped the third and fifth items in the pattern, then Array[0] = 3 and Array[1] = 5.

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetSkippedItemCount Method](#)
[ILinearPatternFeatureData::ISetSkippedItemArray Method](#)
[ILinearPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InstancesToVary Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : InstancesToVary Property (ILinearPatternFeatureData)

Gets or sets whether to vary the spacing of pattern instances in this linear pattern feature.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property InstancesToVary As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.InstancesToVary = value
```

```
value = instance.InstancesToVary
```

C#

```
System.bool InstancesToVary {get; set;}
```

C++/CLI

```
property System.bool InstancesToVary {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to vary the spacing of pattern instances, false to not

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::InstancesToVary](#).

Example

See the [InstanceToVaryOptions](#) examples.

Remarks

This property is valid only when editing the linear pattern feature. You cannot set this property during creation of the feature.

To vary instances in a linear pattern feature:

1. Call [IFeature::GetDefinition](#).
2. Call [ILinearPatternFeatureData::AccessSelections](#).
3. Set this property to true.
4. Call [IFeature::ModifyDefinition](#).
5. Call [ILinearPatternFeatureData::AccessSelections](#).
6. Call [ILinearPatternFeatureData::GetInstanceToVaryOptions](#).
7. Modify [InstanceToVaryOptions](#) properties.
8. Call [ILinearPatternFeatureData::SetInstanceToVaryOptions](#).
9. Call [IFeature::ModifyDefinition](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsDirection2Specified Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : IsDirection2Specified Method (ILinearPatternFeatureData)

Gets whether direction 2 is specified for this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IsDirection2Specified() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Boolean  
  
value = instance.IsDirection2Specified()
```

C#

```
System.bool IsDirection2Specified()
```

C++/CLI

```
System.bool IsDirection2Specified();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if direction 2 is specified, false if not

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::IsDirection2Specified](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetD2AxisType Method](#)
[ILinearPatternFeatureData::D2Axis Property](#)
[ILinearPatternFeatureData::D2PatternSeedOnly Property](#)
[ILinearPatternFeatureData::D2ReverseDirection Property](#)
[ILinearPatternFeatureData::D2Spacing Property](#)
[ILinearPatternFeatureData::D2TotalInstances Property](#)

Availability

SOLIDWORKS 2001Plus SP3, Revision Number 10.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetPatternBodyArray Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : ISetPatternBodyArray Method (ILinearPatternFeatureData)

Sets the seed bodies for this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetPatternBodyArray( _
    ByVal BodyCount As System.Integer, _
    ByRef ArrayDataIn As Body2 _
)
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim BodyCount As System.Integer
Dim ArrayDataIn As Body2

instance.ISetPatternBodyArray(BodyCount, ArrayDataIn)
```

C#

```
void ISetPatternBodyArray(
    System.int BodyCount,
    ref Body2 ArrayDataIn
)
```

C++/CLI

```
void ISetPatternBodyArray(
    & System.int BodyCount,
    & Body2^% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyCount

Number of seed bodies

ArrayDataIn

- In-process, unmanaged C++: Pointer to an array of seed [bodies](#) of size BodyCount

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILinearPatternFeatureData Interface](#)

[ILinearPatternFeatureData Members](#)

[ILinearPatternFeatureData::GetPatternBodyCount Method](#)

[ILinearPatternFeatureData::!GetPatternBodyArray Method](#)

[ILinearPatternFeatureData::PatternBodyArray Property](#)

[ILinearPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetPatternFaceArray Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : ISetPatternFaceArray Method (ILinearPatternFeatureData)

Sets the list of patterned faces for this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetPatternFaceArray( _
    ByVal FaceCount As System.Integer, _
```

```
ByRef ArrayDataIn As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim FaceCount As System.Integer  
Dim ArrayDataIn As System.Object  
  
instance.ISetPatternFaceArray(FaceCount, ArrayDataIn)
```

C#

```
void ISetPatternFaceArray(  
    System.int FaceCount,  
    ref System.object ArrayDataIn  
)
```

C++/CLI

```
void ISetPatternFaceArray(  
& System.int FaceCount,  
& System.Object^% ArrayDataIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceCount

Number of faces

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array of faces of size FaceCount

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILinearPatternFeatureData Interface](#)

[ILinearPatternFeatureData Members](#)

[ILinearPatternFeatureData::GetPatternFaceCount Method](#)

[ILinearPatternFeatureData::IGetPatternFaceArray Method](#)

[ILinearPatternFeatureData::PatternFaceArray Property](#)

[ILinearPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetPatternFeatureArray Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : ISetPatternFeatureArray Method (ILinearPatternFeatureData)

Sets a list of pattern features for this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetPatternFeatureArray( _  
    ByVal FeatCount As System.Integer, _  
    ByRef ArrayDataIn As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim FeatCount As System.Integer
Dim ArrayDataIn As System.Object

instance.ISetPatternFeatureArray(FeatCount, ArrayDataIn)
```

C#

```
void ISetPatternFeatureArray(
    System.int FeatCount,
    ref System.object ArrayDataIn
)
```

C++/CLI

```
void ISetPatternFeatureArray(
    & System.int FeatCount,
    & System.Object^% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatCount

Number of features

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array features of size FeatCount
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetPatternFeatureCount Method](#)
[ILinearPatternFeatureData::GetPatternFeatureCount Method](#)
[ILinearPatternFeatureData::PatternFeatureArray Property](#)
[ILinearPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISetSkippedItemArray Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : ISetSkippedItemArray Method (ILinearPatternFeatureData)

Sets the list of items skipped for this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetSkippedItemArray( _
    ByVal FeatCount As System.Integer, _
    ByRef ArrayDataIn As System.Integer _
)
```

Visual Basic (Usage)

```
Dim instance As ILinearPatternFeatureData
Dim FeatCount As System.Integer
Dim ArrayDataIn As System.Integer
```



```
instance.ISetSkippedItemArray(FeatCount, ArrayDataIn)
```

C#

```
void ISetSkippedItemArray(  
    System.int FeatCount,  
    ref System.int ArrayDataIn  
)
```

C++/CLI

```
void ISetSkippedItemArray(  
& System.int FeatCount,  
& System.int% ArrayDataIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatCount

Number of instances to skip

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array of skipped instances of size FeatCount

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ILinearPatternFeatureData Interface](#)

[ILinearPatternFeatureData Members](#)

[ILinearPatternFeatureData::GetSkippedItemCount Method](#)

[ILinearPatternFeatureData::ISetSkippedItemArray Method](#)

[ILinearPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PatternBodyArray Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : PatternBodyArray Property (ILinearPatternFeatureData)

Gets or sets the seed bodies in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property PatternBodyArray As System.Object
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
```

```
Dim value As System.Object
```

```
instance.PatternBodyArray = value
```

```
value = instance.PatternBodyArray
```

C#

```
System.object PatternBodyArray {get; set;}
```

C++/CLI

```
property System.Object^ PatternBodyArray {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [bodies](#) to pattern

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::PatternBodyArray](#).

Remarks

This property is valid only if:

- you have a multibody part,
- and -
- [ILinearPatternFeatureData::BodyPattern](#) is true.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetPatternBodyCount Method](#)
[ILinearPatternFeatureData::IGetPatternBodyArray Method](#)
[ILinearPatternFeatureData::ISetPatternBodyArray Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PatternElement Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : PatternElement Property (ILinearPatternFeatureData)

Gets or sets the type of entities on which to base this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property PatternElement As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Integer
```

```
instance.PatternElement = value
```

```
value = instance.PatternElement
```

C#

```
System.int PatternElement {get; set;}
```

C++/CLI

```
property System.int PatternElement {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of entities on which to base this linear pattern as defined in [swPatternElementSelection_e](#)

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::PatternElement](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::BodyPattern Property \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

PatternFaceArray Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : PatternFaceArray Property (ILinearPatternFeatureData)

Gets or sets the seed faces for this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property PatternFaceArray As System.Object

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Object
```

```
instance.PatternFaceArray = value
```

```
value = instance.PatternFaceArray
```

C#

```
System.object PatternFaceArray {get; set;}
```

C++/CLI

```
property System.Object^ PatternFaceArray {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [faces](#) to pattern

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::PatternFaceArray](#).

Remarks

This property is valid only if [ILinearPatternFeatureData::BodyPattern](#) is false.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetPatternFaceCount Method](#)
[ILinearPatternFeatureData::IGetPatternFaceArray Method](#)
[ILinearPatternFeatureData::ISetPatternFaceArray Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PatternFeatureArray Property (ILinearPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : PatternFeatureArray Property (ILinearPatternFeatureData)

Gets or sets the seed features in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property PatternFeatureArray As System.Object

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Object
```

```
instance.PatternFeatureArray = value  
value = instance.PatternFeatureArray
```

C#

```
System.object PatternFeatureArray {get; set;}
```

C++/CLI

```
property System.Object^ PatternFeatureArray {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [features](#) for this pattern

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::PatternFeatureArray](#).

Example

[Change Linear Pattern \(C#\)](#)
[Change Linear Pattern \(VB.NET\)](#)
[Change Linear Pattern \(VBA\)](#)

Remarks

This property is valid only if [ILinearPatternFeatureData::BodyPattern](#) is false.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetPatternFeatureCount Method](#)
[ILinearPatternFeatureData::IGetPatternFeatureArray Method](#)
[ILinearPatternFeatureData::ISetPatternFeatureArray Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PropagateVisualProperty Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : PropagateVisualProperty Property (ILinearPatternFeatureData)

Gets or sets whether pattern instances inherit the visual properties (e.g., colors, textures, etc.) of the original seed feature in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property PropagateVisualProperty As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Boolean
```

```
instance.PropagateVisualProperty = value
```

```
value = instance.PropagateVisualProperty
```

C#

```
System.bool PropagateVisualProperty {get; set;}
```

C++/CLI

```
property System.bool PropagateVisualProperty {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to propagate all visual properties, false to not

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::PropagateVisualProperty](#).

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReleaseSelectionAccess Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : ReleaseSelectionAccess Method (ILinearPatternFeatureData)

Releases access to the selections that created this linear pattern.

.NET Syntax

Visual Basic (Declaration)

```
Sub ReleaseSelectionAccess()
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
  
instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::ReleaseSelectionAccess](#).

Remarks

[ILinearPatternFeatureData::AccessSelections](#) and [ILinearPatternFeatureData::!AccessSelections2](#) put the model into a rollback state to allow access to the selections that define this feature.

Use this method to restore the rollback state if you did not modify the feature or use [IFeature::ModifyDefinition](#) or [IFeature::!ModifyDefinition2](#) if you did.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic](#).

SetFeatureScope Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : SetFeatureScope Method (ILinearPatternFeatureData)

Sets the feature scope, whether to autoselect the affected bodies, and the affected bodies in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function SetFeatureScope( _  
    ByVal FeatureScopeOption As System.Boolean, _  
    ByVal AutoSelectBodies As System.Boolean, _  
    ByVal Bodies As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim FeatureScopeOption As System.Boolean  
Dim AutoSelectBodies As System.Boolean  
Dim Bodies As System.Object
```

```
Dim value As System.Boolean
```

```
value = instance.SetFeatureScope(FeatureScopeOption, AutoSelectBodies, Bodies)
```

C#

```
System.bool SetFeatureScope(  
    System.bool FeatureScopeOption,  
    System.bool AutoSelectBodies,  
    System.object Bodies  
)
```

C++/CLI

```
System.bool SetFeatureScope(  
& System.bool FeatureScopeOption,  
& System.bool AutoSelectBodies,  
& System.Object^ Bodies  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatureScopeOption

True to specify affected bodies, false to apply the pattern to all bodies every time the feature regenerates (see **Remarks**)

AutoSelectBodies

True to automatically select all bodies intersected by this pattern feature, false to specify affected bodies (see **Remarks**)

Bodies

Array of [bodies](#) to be affected; valid only if FeatureScopeOption is true and AutoSelectBodies is false

Return Value

True if feature scope set successfully, false if not (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::SetFeatureScope](#).

Remarks

If this method returns false, then default values for FeatureScopeOption and AutoSelectBodies are set. A subsequent call to [IFeatureManager::CreateFeature](#) creates the feature with a FeatureScopeOption of true and an AutoSelectBodies of true.

After calling IFeature::ModifyDefinition, call [IFeature::GetErrorCodes2](#) to determine what's wrong and then take necessary remedial action.

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)

[ILinearPatternFeatureData Members](#)

[ILinearPatternFeatureData::AutoSelect Property \(\)](#)

[ILinearPatternFeatureData::FeatureScope Property \(\)](#)

[ILinearPatternFeatureData::FeatureScopeBodies Property \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetInstanceToVaryOptions Method (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : SetInstanceToVaryOptions Method (ILinearPatternFeatureData)

Sets the options for varying the spacing of pattern instances in this circular pattern feature.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function SetInstanceToVaryOptions( _  
    ByVal InstancesToVaryObj As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim InstancesToVaryObj As System.Object  
Dim value As System.Boolean  
  
value = instance.SetInstanceToVaryOptions(InstancesToVaryObj)
```

C#

```
System.bool SetInstanceToVaryOptions(  
    System.object InstancesToVaryObj  
)
```

C++/CLI

```
System.bool SetInstanceToVaryOptions(  
& System.Object^ InstancesToVaryObj  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[InstancesToVaryObj](#)
[InstanceToVaryOptions](#)

Return Value

True if options successfully set, false if not

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::SetInstanceToVaryOptions](#).

Example

See the [IInstanceToVaryOptions](#) examples.

Remarks

This method is valid only if [ILinearPatternFeatureData::InstancesToVary](#) is true.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetInstanceToVaryOptions Method \(\)](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

SOLIDWORKS API Help

[Send comments on this topic.](#)

SkippedItemArray Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : SkippedItemArray Property (ILinearPatternFeatureData)

Gets or sets the items skipped in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property SkippedItemArray As System.Object

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData  
Dim value As System.Object
```

```
instance.SkippedItemArray = value
```

```
value = instance.SkippedItemArray
```

C#

```
System.object SkippedItemArray {get; set;}
```

C++/CLI

```
property System.Object^ SkippedItemArray {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of integers representing skipped items

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::SkippedItemArray](#).

Remarks

The array is 0-based. If you skip the third and fifth items, then the array looks like ArrayOut[0]=3 and ArrayOut[1]=5.

To calculate the pattern instance numbers to skip:

Calculate:

For a bi-directional pattern:

$$I = n2 * (i - 1) + (j - 1)$$

For a uni-directional pattern:

$$I = i - 1$$

where:

- I = pattern instance number
- $n2$ = number of instances in Direction 2
- i = index for Direction 1
- j = index for Direction 2

In the pattern's PropertyManager, find $n2$ in the **Direction 2 > Spacing and Instances > Number of instances** field and find $[i,j]$ in the **Modified Instances** section.

See [Accessing Selections that Define Features](#) for additional details on using this property.

For more information, see the **Linear Patterns and the Linear Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)
[ILinearPatternFeatureData::GetSkippedItemCount Method](#)
[ILinearPatternFeatureData::IGetSkippedItemArray Method](#)
[ILinearPatternFeatureData::ISetSkippedItemArray Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : StructureSystemToPatternArray Property (ILinearPatternFeatureData)

Gets or sets the structure systems to pattern in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property StructureSystemToPatternArray As System.Object

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Object

instance.StructureSystemToPatternArray = value

value = instance.StructureSystemToPatternArray
```

C#

```
System.object StructureSystemToPatternArray {get; set;}
```

C++/CLI

```
property System.Object^ StructureSystemToPatternArray {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [IStructureSystemFolders](#)

Visual Basic for Applications (VBA) Syntax

See [LinearPatternFeatureData::StructureSystemToPatternArray](#).

Example

'VBA

'This example demonstrates how to create a linear pattern using a structure system.

'=====

'Preconditions:

- '1. Ensure the specified part template exists.
- '2. Open an Immediate window.
- '3. Press F5.

'

'Postconditions:

- '1. Creates Sketch1 containing two sketch segments.
- '2. Configures the start/end extensions and the member profile.
- '3. Selects the two sketch segments.
- '4. Creates primary **Structure System1** with two primary path segment members (**Member1** and **Member2**).
- '5. Creates a **Boss-Extrude1** to pattern.
- '6. Inspect the graphics area.
- '7. Press F5 to create **LPattern1**.
- '8. Press F5 to finish.
- '9. Inspect the Immediate window.

'=====

```
Dim swApp As SldWorks.SldWorks
Dim modDoc As SldWorks.ModelDoc2
Dim swFeatMgr As SldWorks.FeatureManager
Dim swSelMgr As SldWorks.SelectionMgr
Dim modDocExt As SldWorks.ModelDocExtension
Dim structMemDef As SldWorks.StructureSystemMemberFeatureData
Dim profDef As SldWorks.StructureSystemMemberProfile
Dim PrimDef As SldWorks.PrimaryStructuralMemberFeatureData
Dim memPathSegDef As PrimaryMemberPathSegmentFeatureData
Dim memBetweenPointsSecDef As SldWorks.SecondaryMemberBetweenPointsFeatureData
Dim secMemDatTochange As SldWorks.SecondaryMemberBetweenPointsFeatureData
```

```

Dim memSupportPlaneSecDef As SldWorks.SecondaryMemberSupportPlaneFeatureData
Dim segments(2) As Object
Dim stat As Boolean
Dim memData(0) As SldWorks.StructureSystemMemberFeatureData
Dim memDataArray As Variant
Dim structSys As SldWorks.Feature
Dim baseSecondarySystem As SldWorks.StructureSystemFolder
Dim structSysDef As SldWorks.StructureSystemFolder
Dim structSysSecDef As SldWorks.StructureSystemFolder
Dim feat As SldWorks.Feature
Dim structSysModDef As SldWorks.StructureSystemFolder
Dim outProfiles As Variant
Dim MemberData As SldWorks.StructureSystemMemberFeatureData
Dim memberArray As Variant
Dim profileGrpFeat As SldWorks.Feature
Dim profileGrp As SldWorks.ProfileGroupFolder
Dim memToChange As SldWorks.StructureSystemMemberFeatureData
Dim i As Long
Dim j As Long
Dim secDef As SldWorks.SecondaryStructuralMemberFeatureData
Dim UpToMemDef As SldWorks.SecondaryMemberUpToMembersFeatureData
Dim point As Object
Dim Members(0) As Object

Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swSketchMgr As SldWorks.SketchManager
Dim swFeatureMgr As SldWorks.FeatureManager
Dim swSelectionMgr As SldWorks.SelectionMgr
Dim swSketchSegment As SldWorks.SketchSegment
Dim swFeature As SldWorks.Feature
Dim swLinearPatternFeatureData As SldWorks.LinearPatternFeatureData
Dim sketchSegments As Variant
Dim status As Boolean
Dim obj As Object
Dim patternFeatures(1) As Object

Option Explicit

Sub main()
    Set swApp = Application.SldWorks

    Dim swSheetWidth As Double
    swSheetWidth = 0
    Dim swSheetHeight As Double
    swSheetHeight = 0
    Set modDoc = swApp.NewDocument("C:\ProgramData\SolidWorks\SOLIDWORKS 2023\templates\Part.PRTDOT", 0, swSheetWidth,
swSheetHeight)

    Set modDocExt = modDoc.Extension

    modDoc.SketchManager.InsertSketch True
    stat = modDocExt.SelectByID2("Front Plane", "PLANE", -0.077188654347454, 0.054268560279924, 3.86214196026222E-03, False, 0,
Nothing, 0)
    modDoc.ClearSelection2 True

    Dim skSegment As Object
    Set skSegment = modDoc.SketchManager.CreateLine(-0.168061, 0.084736, 0#, -0.168061, -0.077684, 0#)
    Set skSegment = modDoc.SketchManager.CreateLine(0.075216, 0.107771, 0#, 0.075216, -0.006699, 0#)
    modDoc.ClearSelection2 True

    stat = modDocExt.SelectByID2("Top Plane", "PLANE", 0, 0, 0, True, 0, Nothing, 0)
    Dim myRefPlane As SldWorks.RefPlane
    Set myRefPlane = modDoc.FeatureManager.InsertRefPlane(8, 0.03, 0, 0, 0, 0)
    modDoc.ClearSelection2 True
    stat = modDocExt.SelectByID2("Top Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
    modDoc.ClearSelection2 True
    modDoc.SketchManager.InsertSketch True

    stat = modDocExt.SelectByID2("Line1@Sketch1", "EXTSKETCHSEGMENT", -0.168061313304597, 5.44142573846353E-02, 0, False, 0,
Nothing, 0)
    stat = modDocExt.SelectByID2("Line2@Sketch1", "EXTSKETCHSEGMENT", 7.52162521083513E-02, 5.32390034454423E-02, 0, True, 0,
Nothing, 0)

    'Create primary structure system path segment members
    Set structMemDef = modDocExt.CreateStructureSystemMemberData(0)
    Debug.Print "Type of structure system member as defined by swStructureSystemMemberType_e: " &
structMemDef.StructureSystemMemberType

    structMemDef.StartEndExtendD1 = 1#
    structMemDef.StartEndExtendD2 = 2#

```

```

Set profDef = structMemDef.MemberProfile

profDef.ProfileStandard = "ansi inch"
profDef.ProfileType = "c channel"
profDef.ProfileSize = "3 x 5"

Set PrimDef = structMemDef
Debug.Print "Structure system primary member creation type as defined by swStructureSystemMemberCreationType_e: " &
PrimDef.PrimaryMemberType

Set memPathSegDef = PrimDef
memPathSegDef.MergeTangentMembers = True

Set swSelMgr = modDoc.SelectionManager

Dim segments(1) As Object
Set segments(0) = swSelMgr.GetSelectedObject6(1, 0)
Set segments(1) = swSelMgr.GetSelectedObject6(2, 0)
stat = memPathSegDef.SetPathSegments(segments)
Debug.Print "Path segments successfully set: " & stat

Dim PrimMemData(0) As SldWorks.StructureSystemMemberFeatureData
Set PrimMemData(0) = structMemDef

Dim PrimMemDatArray As Variant
PrimMemDatArray = PrimMemData

Dim structSysFeat As SldWorks.Feature
Set structSysFeat = modDocExt.CreateStructureSystem(PrimMemDatArray, Nothing)

'Create the axis for the circular pattern
'status = modDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
'modDoc.SketchManager.InsertSketch True
'modDoc.ClearSelection2 True
'Dim skSegmentAxis As Object
'Set skSegmentAxis = modDoc.SketchManager.CreateLine(0.695357, 0.586267, 0#, 0.695357, -0.704287, 0#)
'modDoc.ClearSelection2 True
'modDoc.SketchManager.InsertSketch True

'Create a solid body to also pattern
status = modDocExt.SelectByID2("Front Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
modDoc.SketchManager.InsertSketch True
modDoc.ClearSelection2 True
status = modDocExt.SketchBoxSelect("0.509957", "-0.990370", "0.000000", "1.024907", "-1.314598", "0.000000")
status = modDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstToRectEntity,
swUserPreferenceOption_e.swDetailingNoOptionSpecified, False)
status = modDocExt.SetUserPreferenceToggle(swUserPreferenceToggle_e.swSketchAddConstLineDiagonalType,
swUserPreferenceOption_e.swDetailingNoOptionSpecified, True)
Dim vSkLines As Variant
vSkLines = modDoc.SketchManager.CreateCornerRectangle(0.713394086918119, -0.907724114572968, 0, 1.44449597678115,
-1.33367043301491, 0)
modDoc.ClearSelection2 True
modDoc.SketchManager.InsertSketch True
modDoc.SketchManager.InsertSketch True
Dim myFeature As SldWorks.Feature
Set myFeature = modDoc.FeatureManager.FeatureExtrusion2(True, False, False, 0, 0, 0.07, 0.01, False, False, False,
1.74532925199433E-02, 1.74532925199433E-02, False, False, False, False, False, True, True, 0, 0, False)
modDoc.SelectionManager.EnableContourSelection = False

modDoc.ViewZoomtofit

Stop

Set swModel = swApp.ActiveDoc
Set swModelDocExt = swModel.Extension
Set swSketchMgr = swModel.SketchManager
Set swFeatureMgr = swModel.FeatureManager
Set swSelectionMgr = swModel.SelectionManager

'Create linear pattern
Set swLinearPatternFeatureData = swFeatureMgr.CreateDefinition(swFmLPattern)
swLinearPatternFeatureData.D1EndCondition = 0
swLinearPatternFeatureData.D1ReverseDirection = True
swLinearPatternFeatureData.D1Spacing = 0.5
swLinearPatternFeatureData.D1TotalInstances = 3
swLinearPatternFeatureData.BodyPattern = True
swLinearPatternFeatureData.GeometryPattern = False
swLinearPatternFeatureData.VarySketch = False

'Preselect direction, bodies, structure system folders
swModel.ClearSelection2 True

```

```

status = swModelDocExt.SelectByRay(1.15634621903263, -1.3336704330149, 6.9999999999932E-02, 0, 0, -1, 2.16151863089938E-02, 1,
False, 1, 0)
status = swModelDocExt.SelectByID2("Boss-Extrude1", "SOLIDBODY", 0, 0, 0, True, 256, Nothing, 0)
status = swModelDocExt.SelectByID2("Structure System1", "BODYFEATURE", 0, 0, 0, True, 134217728, Nothing, 0)

Set swFeature = swFeatureMgr.CreateFeature(swLinearPatternFeatureData)

Stop 'Examine the graphics area

swModel.ClearSelection2 True
status = swModelDocExt.SelectByID2("LPattern1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set swFeature = swSelectionMgr.GetSelectedObject6(1, -1)
Set swLinearPatternFeatureData = swFeature.GetDefinition
swLinearPatternFeatureData.AccessSelections swModel, Nothing
Debug.Print "Is LPattern1 a features/faces pattern or a bodies pattern (true if a bodies pattern)? " & swLinearPatternFeatureData.BodyPatter

Dim var As Variant
var = swLinearPatternFeatureData.StructureSystemToPatternArray

Dim strctSysFolder As StructureSystemFolder
Dim index As Integer
For index = LBound(var) To UBound(var)
    Set strctSysFolder = var(index)
    Set patternFeatures(0) = strctSysFolder
    Debug.Print "Number of structure system folders: " & strctSysFolder.GetProfileGroupFoldersCount
Next

'Uncomment the following lines of code to modify the linear pattern
'to use another structure system (e.g., Structure System2)
'Stop

'Dim feat As Feature
'swModel.ClearSelection2 True
'status = swModelDocExt.SelectByID2("Structure System2", "BODYFEATURE", 0, 0, 0, True, 0, Nothing, 0)
'Set feat = swSelectionMgr.GetSelectedObject6(1, 0)
'Set patternFeatures(1) = feat.GetSpecificFeature2()

"During linear pattern modification, use StructureSystemPatternArray property to change structure system to pattern
'swLinearPatternFeatureData.StructureSystemToPatternArray = patternFeatures

Debug.Print swFeature.ModifyDefinition(swLinearPatternFeatureData, swModel, Nothing)

End Sub

```

Remarks

This property is valid only if [ILinearPatternFeatureData.BodyPattern](#) is true.

Use the setter of this property to change structure systems only after the linear pattern feature has been created.

Before using this property in .NET code, read [Dispatch Object Arrays as Input in .NET](#). To remove a structure system from the pattern, set this property to a Dispatcher array containing one null element.

For more information, see the **SOLIDWORKS Design user-interface help > Weldments and Structure System > Pattern and Mirror Support** topic.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments on this topic.](#)

VarySketch Property (ILinearPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ILinearPatternFeatureData Interface](#) : VarySketch Property (ILinearPatternFeatureData)

Gets or sets whether to allow the pattern instances of a seed sketch to vary in relation to enclosing geometry in this linear pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property VarySketch As System.Boolean

Visual Basic (Usage)



```
Dim instance As ILinearPatternFeatureData
Dim value As System.Boolean

instance.VarySketch = value

value = instance.VarySketch
```

C#

```
System.bool VarySketch {get; set;}
```

C++/CLI

```
property System.bool VarySketch {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to allow the pattern instances of a seed sketch to vary in relation to enclosing geometry, false to not

Visual Basic for Applications (VBA) Syntax

See [ILinearPatternFeatureData::VarySketch](#).

Example

See the [ILinearPatternFeatureData](#) example.

Remarks

This property is valid only when creating a pattern from a seed sketch.

The seed sketch has:

- An enclosing boundary of base geometry.
- Fully defined relations.
- Dimensions only for measurements that will not vary in the pattern instances.
- A driving dimension that is used to specify the direction of the linear pattern.

If this property is true:

- Specify [ILinearPatternFeatureData::D1Axis](#) with a driving dimension of the seed sketch.
- Pattern instances vary with and are constrained by the enclosing geometry.

See the **SOLIDWORKS Help > Parts and Features > Features > Patterns and Mirroring > Control and Modify Patterns > Vary Sketch** topic.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ILinearPatternFeatureData Interface](#)
[ILinearPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0